

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 | SUB-STRAND 2.1: SIMILARITY AND ENLARGEMENT | FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How can a full-size photograph of a person be resized to fit a small passport photo box without distortion? Remember Wanjiru Atieno of Nyalenda Secondary School, Kisumu, and her four prints from Mr Otieno at Lakeview Photo Studio: the original 14 cm × 18 cm, the passport photo 3.5 cm × 4.5 cm, the prize-giving poster 70 cm × 90 cm, and the 4.5 cm × 4.5 cm trial print made by changing only the width. Answer each section in the space under it. Use your own measurements and calculations from Lessons 1 to 12. Write the claim first, then evidence (numbers), then reasoning (why the numbers matter). Use k for the linear scale factor, $k = \text{image length} \div \text{object length}$, and always say which print is the object and which is the image. Write width first, then height. Show your working and include units.

Section 1: Claim and evidence from the ratios and angles

Use the table below, which comes from your measurements of the four prints.

Print	Width × height (cm)	Width : height	Chin angle
Original	14 × 18	7 : 9	about 42.5°
Passport	3.5 × 4.5	7 : 9	about 42.5°
Poster	70 × 90	7 : 9	about 42.5°
Trial print	4.5 × 4.5	1 : 1	about 53°

(a) Write a claim: which prints keep Wanjiru's likeness, and which print does not? (b) Use the width : height ratios and the chin angles to give evidence for your claim. (c) Explain, using the words similar (~), corresponding sides and corresponding angles, what must stay the same when a picture changes size.

Write your answer here:

Section 2: Scale factors

Take the original $14\text{ cm} \times 18\text{ cm}$ print as the object. (a) Calculate the linear scale factor k for the passport photo (image) for width and for height. State whether this is an enlargement or a reduction. (b) Calculate k for the poster (image), for width and for height. (c) Find k from the passport photo (object) to the poster (image). (d) For the $4.5\text{ cm} \times 4.5\text{ cm}$ trial print (image), calculate the width factor and the height factor from the original. Compare them and explain what your comparison tells you about the face in this print.

Write your answer here:

Section 3: Using scale factor and similar triangles to find unknown lengths

(a) In the original print the face triangle has an eye gap of 5.6 cm and a chin distance of 7.2 cm. Use k from the original to the passport photo to find the eye gap and the chin distance on the passport photo. (b) Now find the eye gap and the chin distance on the poster. (c) At Nyalenda Secondary School, at the same time of day, Wanjiru (1.5 m tall) casts a 2 m shadow and a tree casts a 12 m shadow. Explain why the two shadow triangles can be treated as similar, then find the height of the tree. (d) Explain how this problem is related to Wanjiru's photo.

Write your answer here:

Section 4: Area and volume factors

(a) Calculate the area of the original print, the passport photo and the poster. **(b)** Use your areas and the linear scale factors to find the area factor from the original to the passport photo and from the original to the poster. Write the pattern that connects the area factor to k . **(c)** Mr Otieno packs prints in a small carton of $10\text{ cm} \times 6\text{ cm} \times 4\text{ cm}$ and a larger similar carton of $20\text{ cm} \times 12\text{ cm} \times 8\text{ cm}$. Find k , the volumes of both cartons and the volume factor. Write the pattern that connects the volume factor to k . **(d)** A friend says: 'If every length of the poster is multiplied by 5, then its area is multiplied by 5.' Explain whether your friend is correct.

Write your answer here:

Section 5: Enlargement, applications and the final explanation

(a) Triangle ABC has A(1,1), B(3,1) and C(1,2). Write the image coordinates for an enlargement with centre O(0,0) and scale factor 2. Explain how the rays from O are used to build the image. (b) Triangle PQR with P(2,2), Q(3,2), R(2,3) has the image P'(4,4), Q'(7,4), R'(4,7). Explain how to find the centre of the enlargement and find k. (c) On a 1:5 000 000 map, Kisumu and Nairobi are about 265 km apart in a straight line. Find their distance on the map in cm. (d) A school sign of 70 cm × 90 cm (0.63 m²) is enlarged with k = 3. Paint costs KES 400 per m². Find the cost of paint for each sign. (e) Now answer the driving question: How can a full-size photograph be resized to fit a small passport photo box without distortion? Also say why we need to enlarge or reduce objects, and how we use similarity and enlargement in day-to-day life.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim, evidence and reasoning	States a clear claim, supports it with exact data (7:9 vs 1:1, about 42.5° vs about 53° , $k = 1/4$ and 5 vs $9/28$ and $1/4$) and links the evidence to the idea of similarity.	States a correct claim and uses some correct data, but the reasoning linking the data to similarity is incomplete.	Claim is unclear or missing, or uses little or incorrect evidence.
Scale factor calculations	Calculates every k as $\text{image} \div \text{object}$, names the object and the image, and correctly identifies enlargement or reduction. Compares the unequal factors of the trial print.	Calculates most scale factors correctly but sometimes omits the object and image or the enlargement/reduction label.	Calculations are mostly incorrect or the object and image are mixed up.
Area and volume factors	Correctly uses k^2 and k^3 with Wanjiru's prints and the cartons, and explains why doubling lengths does not double the area.	Uses k^2 and k^3 correctly in most cases but explanations are brief.	Treats area or volume as changing by k , or makes errors in the calculations.
Enlargement, applications and mathematical language	Finds images and the centre correctly, solves the map, sign cost and shadow problems accurately, and uses $\sim$, corresponding sides and corresponding angles, units and width-first order	Solves most application problems correctly with minor errors in units, notation or the order of width and height.	Has many errors in the applications, or little correct use of terms and units.

	correctly.		
Answering the driving question	Gives a complete explanation that size can change but shape is kept only if one scale factor is applied to all lengths. Connects to the Key Inquiry Questions with real examples.	Gives a correct answer to the driving question but with limited links to everyday examples.	Answer is incomplete or does not explain why the trial print is distorted.

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